

A Data-Driven Framework for Product-Line Profitability, Margin Risk, and Portfolio Decision Support: A Case Study of Nassau Candy Distributor

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Abstract—This paper presents an end-to-end business analytics framework for evaluating product-line profitability, margin quality, concentration risk, cost structure, and margin stability in a distributor portfolio. The empirical study uses 10,194 cleaned Nassau Candy Distributor transaction records covering 8,549 unique orders, 5,044 customers, 15 products, three divisions, four regions, and 59 states/provinces. The analysis integrates gross-margin and profit-per-unit metrics with product ranking, division-level revenue–profit comparison, Pareto concentration, geographic dependency, portfolio-relative cost diagnostics, and margin-volatility assessment. A methodological feature is the use of distribution-based thresholds for cost diagnostics: the first quartile of gross margin, third quartile of cost ratio, first quartiles of sales and gross profit, and median sales are used to identify cost-heavy, margin-poor, repricing, cost-renegotiation, and discontinuation-review candidates. Results show that the company generated \$141,783.63 in sales and \$93,442.80 in gross profit, corresponding to an overall gross margin of 65.91%. Five Chocolate products account for 92.88% of revenue and 95.06% of gross profit, revealing substantial portfolio dependence. Kazookles is the clearest pricing/cost-risk product, with a 92.31% cost ratio and 7.69% gross margin, while Fun Dip, SweetARTS, and Nerds meet the portfolio-relative criteria for discontinuation review. Geographic results show that 16 of 59 states/provinces account for approximately 80% of revenue and profit. The framework is operationalized through an interactive Streamlit dashboard with date, division, margin-threshold, and product-search controls. The study demonstrates how EDA, financial KPIs, concentration analysis, and interactive visualization can be combined into a practical product-portfolio decision-support system.

Index Terms—business analytics, profitability analysis, gross margin, Pareto analysis, cost diagnostics, margin volatility, portfolio concentration, Streamlit dashboard, exploratory data analysis.

I. INTRODUCTION

Distributor performance is often evaluated using revenue growth, order volume, or unit movement. These measures are useful, but they do not reveal whether sales create economic value. A product can generate substantial revenue while absorbing most of that revenue in cost, and a high-margin product can be financially immaterial when its sales base is extremely small. Effective portfolio management therefore requires a joint view of scale, unit economics, contribution, concentration, and stability. Exploratory data analysis (EDA) provides a disciplined approach for discovering such structure before predictive or prescriptive modeling is attempted [1].

The Nassau Candy Distributor project addresses this problem through a staged analytical workflow designed around product-line profitability and margin performance. Rather than treating the dataset as a generic visualization exercise, the project converts transaction-level order records into managerial signals: gross-margin leadership, profit contribution, division efficiency, portfolio dependency, geographic concentration, cost pressure, and margin volatility. The resulting analysis is intended to answer both descriptive questions—what happened and where profit is generated—and diagnostic questions—why some products create weak economics despite sales activity.

The study contributes a practical decision-support framework with four linked characteristics. First, it combines financial KPIs and EDA instead of relying on sales rankings alone. Second, it applies Pareto-style concentration analysis to both products and geography to identify dependency risk. Third, it uses empirical quantiles from the observed portfolio to define cost and margin risk rather than imposing an arbitrary universal cutoff. Fourth, it translates the analytical outputs into an interactive Streamlit dashboard so that managers can filter by date, division, margin threshold, and product name. This combination reflects the broader principle that data products create value when analytical

methods are tied to operational decisions rather than presented as isolated charts [2].

A. Research Objectives and Questions

The principal objective is to establish a reproducible product-portfolio analytics process that identifies sources of profit, structural weaknesses, and concentration risks. The analysis is organized around the following research questions (RQs).

- RQ1: Which products and divisions create the greatest gross profit and margin, and do high-sales products consistently translate into high profit?
- RQ2: How concentrated are revenue and gross profit across products, states/provinces, and regions?
- RQ3: Which products exhibit a high cost burden and weak margin relative to the observed portfolio distribution?
- RQ4: Which products should be prioritized for repricing, cost renegotiation, or discontinuation review under a transparent rule set?
- RQ5: How stable are company, division, and product margins in the supplied time-based analysis?
- RQ6: How can the analytical results be operationalized in an interactive dashboard for business users?

B. Main Contributions

The main contribution is not a new statistical estimator; it is an integrated analytical design for portfolio decision support. The paper demonstrates how standard financial formulas, EDA, empirical quantiles, concentration analysis, and interactive visualization can be assembled into a transparent workflow. In an applied setting, this transparency is important because managers must be able to understand why a product has been flagged and what evidence supports the recommended action.

II. ANALYTICAL CONTEXT AND RELATED WORK

EDA emphasizes graphical and numerical techniques for understanding the structure of data before formal modeling [1]. In

business analytics, this philosophy is particularly useful because operational datasets frequently contain data-quality problems, skewed distributions, aggregation effects, and business definitions that must be validated before higher-level conclusions are made. The present work follows that approach by beginning with data profiling, validation, and reconciliation before product and division comparisons are performed.

Profitability analysis connects sales outcomes to the resources consumed to generate them. Cost-management literature stresses that absolute revenue or cost levels are insufficient without understanding the relationship between cost, margin, and product economics [3]. The Nassau Candy framework therefore evaluates gross margin, profit per unit, cost ratio, and contribution alongside total sales. This avoids a common interpretation error in which a high-cost product is automatically labeled inefficient even when its cost is high only because its scale is high.

Concentration analysis is motivated by the Pareto principle and quality-management practice, where a relatively small subset of causes often accounts for a large share of observed effects [4]. In the present study, cumulative revenue and gross-profit shares are used to determine how many products are required to cross an 80% contribution threshold. The same logic is extended to states/provinces and regions. The resulting concentration indicators should be interpreted as dependency measures rather than as causal risk estimates: a concentrated portfolio can be profitable, but it is more exposed to disruptions affecting its dominant products or markets.

Interactive dashboards complement static analysis by allowing a user to change the analytical scope and immediately observe the consequences. Dashboard-design guidance emphasizes focus, context, and the avoidance of unnecessary visual clutter [5]. The implementation in this project uses a small set of high-value views aligned with the technical requirements: product margin leadership and contribution, division performance, cost-versus-margin diagnostics, Pareto concentration, and volatility. The implementation relies on Python data-processing workflows commonly supported by pandas [6], with charts generated in Matplotlib/Plotly-style analytical workflows and a Streamlit front end [7], [8].

A notable methodological choice is the use of quantiles for cost-diagnostic thresholds. Quartile-based thresholds are not claimed to represent an industry benchmark. Instead, they provide a transparent relative classification within the observed 15-product portfolio. This is appropriate when external benchmark data are unavailable but management still needs a consistent way to prioritize products for review. The paper therefore distinguishes between portfolio-relative flags and universal financial standards.

III. RESEARCH DESIGN AND DATASET

A. Dataset Profile

The analysis uses the cleaned Nassau Candy Distributor transaction dataset and phase-wise analytical reports supplied with the project [9]–[11]. The cleaned dataset contains 10,194 rows and 27 fields, of which 18 are original business fields and nine are validation/quality fields. The transaction period spans 2 January 2024 through 31 December 2025 based on Order Date. The business dimensions cover products, divisions, customers, geographic locations, and order/ship attributes. Financial fields include Sales, Units, Gross Profit, and Cost.

TABLE I. DATASET PROFILE

Measure	Value
Rows	10,194
Unique orders	8,549
Unique customers	5,044
Products	15
Divisions	3
Regions	4
States/provinces	59
Cities	542
Total units	38,654
Total sales	\$141,783.63
Total gross profit	\$93,442.80
Total cost	\$48,340.83

B. Data Quality and Validation

The phase-2 data-quality report contains zero missing values across the 27 fields and the phase-1 report records zero exact duplicate rows in the cleaned analytical extract. Validation flags for sales, units, cost, profit, and profit calculation are present in the dataset. These checks support the use of the financial fields for profitability calculations. However, a separate shipping-date anomaly remains in the archived cleaned file: Shipping Days range from 904 to 1,642 days even though all rows are marked valid. Order dates span 2024–2025, while ship dates extend from 2026 to 2030. Because those values are not credible as operational fulfillment durations, Ship Date and Shipping Days are excluded from profitability and operational conclusions in this paper.

TABLE II. DATA-QUALITY SUMMARY

Validation area	Observed result
Missing values	0 across reported fields
Exact duplicates	0 in phase-1 cleaned report
Sales / units / cost / profit flags	All rows marked valid in archived extract
Order Date	2024-01-02 to 2025-12-31
Shipping Days	904 to 1,642 (implausible)
Treatment	Exclude ship-duration measures from findings

C. Unit of Analysis and Aggregation

The source data are transaction-level records, while most research questions require product-, division-, geographic-, or month-level aggregation. Product metrics are generated by grouping on Product ID, Product Name, and Division. Division metrics aggregate sales, units, cost, and gross profit by Division. Geographic metrics use unique Order ID counts plus units, sales, and gross profit by State/Province and Region. Time-based margin metrics use Order Date rather than Ship Date because of the shipping anomaly described above.

Analytical Workflow



Fig. 1. Analytical workflow from transaction validation to interactive decision support.

IV. METHODOLOGY

A. Core Profitability Metrics

The analysis uses gross profit and sales to establish margin quality, and units to measure product-level profit intensity. Contribution measures place each product in the context of the full portfolio. Let S_i , C_i , G_i , and U_i denote sales, cost, gross profit, and units for product i . The following metrics are used throughout the study.

$$\text{Gross Margin}_i (\%) = (G_i / S_i) \times 100$$

$$\text{Profit per Unit}_i = G_i / U_i$$

$$\text{Revenue Contribution}_i (\%) = (S_i / \sum S_i) \times 100$$

$$\text{Profit Contribution}_i (\%) = (G_i / \sum G_i) \times 100$$

$$\text{Cost Ratio}_i (\%) = (C_i / S_i) \times 100$$

Gross margin evaluates how much of each sales dollar remains after product cost, while profit per unit identifies products that generate strong or weak economics at the unit level. Revenue and profit contribution are particularly important for the Nassau portfolio because many Sugar products have attractive percentage margins but negligible business scale.

B. Product-Level Classification

The project reports product rankings by gross profit and gross margin and assigns portfolio categories such as High Profit High Margin, High Sales Low Margin, and Low Sales Low Profit. These categories are comparative rather than absolute. Therefore, a product such as Everlasting Gobstopper can appear in a high-margin category despite producing only \$104 of gross profit, while large Chocolate products dominate absolute profit. The paper retains the supplied classifications but interprets them together with sales scale and contribution.

C. Division-Level Analysis

Division performance is measured through total sales, total gross profit, division gross margin, profit per unit, revenue contribution, and profit contribution. A useful diagnostic is the difference between revenue share and profit share. When profit share exceeds revenue share, the division is creating more profit than its sales weight would imply; when profit share is lower, the division is economically dilutive relative to its sales contribution.

D. Pareto and Dependency Analysis

Products are sorted in descending order of revenue and gross profit, after which cumulative shares are calculated. The minimum number of products required to cross 80% cumulative contribution is recorded. A smaller percentage of products needed to cross this threshold implies greater portfolio concentration. The same concept is applied to states/provinces. Regional analysis compares order, unit, revenue, and profit shares. These are dependency indicators; they do not by themselves prove operational congestion, capacity limitations, or causal exposure.

E. Quantile-Based Cost Structure Diagnostics

The final cost-diagnostic methodology uses empirical thresholds computed from the 15-product cost table. This corrects the earlier provisional idea of using a fixed 50% threshold. The approved thresholds are: Q1 gross margin = 48.33%, Q3 cost ratio = 51.67%, Q1 sales = \$69.00, median sales = \$597.50, and Q1 gross profit = \$40.36. These values are recomputed from the supplied Cost Structure Diagnostics report.

TABLE III. PORTFOLIO-RELATIVE COST DIAGNOSTIC THRESHOLDS

Threshold	Definition	Observed value
Low margin	Q1 of Gross_Margin(%)	48.33%
High cost ratio	Q3 of Cost_Ratio(%)	51.67%
Low sales	Q1 of Total_Sales	\$69.00
Median sales	Q2 of Total_Sales	\$597.50
Low profit	Q1 of Total_Gross_Profit	\$40.36

A product is cost-heavy when its cost ratio is at or above Q3 and margin-poor when its gross margin is at or below Q1. A pricing inefficiency/repricing candidate must be both cost-heavy and margin-poor while having sales at or above the median, which prevents very small products from being labeled repricing priorities solely because of poor unit economics. Cost renegotiation is triggered by the cost-heavy and margin-poor combination. Discontinuation review requires simultaneous cost-heavy, margin-poor, low-sales, and low-profit conditions. The rules are intended to prioritize review, not automate final managerial decisions.

F. Margin Volatility

Margin volatility is summarized using the standard deviation of periodic gross-margin observations. The supplied volatility report aggregates observations into 12 calendar month-of-year buckets across the 2024–2025 study period rather than 24 separate chronological months. Consequently, the reported company standard deviation of 0.31 percentage points should be interpreted as seasonal month-of-year variation in the supplied report. Product-level margins are effectively constant because product price-cost relationships in the dataset are fixed. Division-level volatility remains informative because monthly product mix changes the aggregate division margin.

$$\text{Margin Volatility} = SD(\text{Margin}_t), \text{ where } t \text{ indexes the supplied monthly buckets}$$

G. Interactive Dashboard

The Streamlit dashboard operationalizes the analytics with four modules specified in the project technical documentation: Product Profitability Overview, Division Performance, Cost vs Margin Diagnostics, and Profit Concentration Analysis [9]. User controls include date range, division filter, margin threshold slider, and product search. Live charts are recalculated from the cleaned row-level dataset, while validated Colab outputs are retained as baseline reports and downloadable evidence.

V. RESULTS

A. Overall Financial Performance

Across the cleaned portfolio, total sales equal \$141,783.63, total cost equals \$48,340.83, and total gross profit equals \$93,442.80. The aggregate gross margin is therefore 65.91%. At the company level this appears financially strong, but the aggregate masks extreme differences across products and divisions. The concentration analyses below show that the strong headline performance is driven overwhelmingly by a small group of Chocolate products.

B. Product Profitability

Scrumdiddlyumptious ranks first in gross profit at \$19,357.50, followed by Triple Dazzle Caramel at \$18,610.20, Milk Chocolate at \$17,443.37, Nutty Crunch Surprise at \$16,819.95, and Fudge MalloWS at \$16,593.60. These five products are all in the Chocolate division and account for nearly the entire profit base. Triple Dazzle Caramel has the highest sales at \$28,485.00.

Nutty Crunch Surprise produces the highest gross margin among the five core Chocolate products at 71.35%.

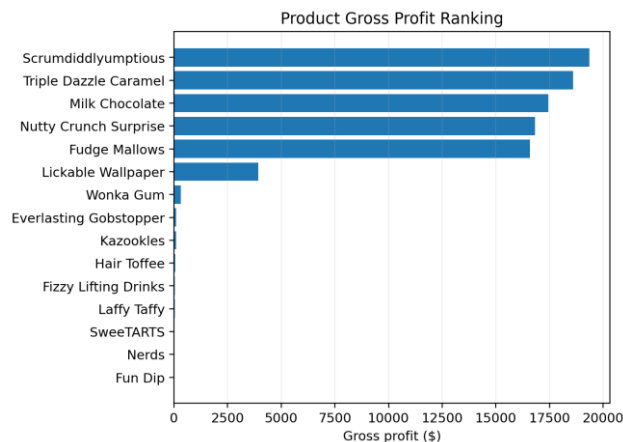


Fig. 2. Product gross-profit ranking.

Margin percentage alone produces a different ordering. Everlasting Gobstopper reports the highest margin at 80.00%, followed by Hair Toffee at 77.78%. Yet their sales are only \$130.00 and \$76.50, respectively. This illustrates the distinction between percentage efficiency and total economic contribution. By contrast, Kazookles records \$1,205.75 in sales but only \$92.75 of gross profit, a 7.69% margin and \$0.25 profit per unit. Its weakness is therefore economically more important than the low sales of several Sugar products.

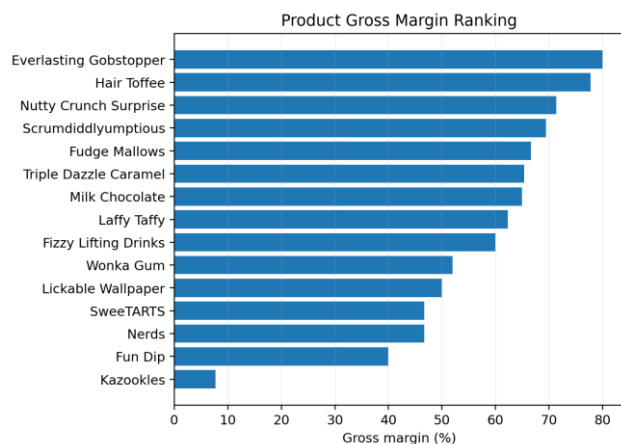


Fig. 3. Product gross-margin ranking.

TABLE IV. SELECTED PRODUCT PROFITABILITY RESULTS

Product	Gross profit (\$)	Margin
Scrumdiddlyumptious	19,357.50	69.44%
Triple Dazzle Caramel	18,610.20	65.33%
Milk Chocolate	17,443.37	64.92%
Nutty Crunch Surprise	16,819.95	71.35%
Fudge Mallows	16,593.60	66.67%
Kazookles	92.75	7.69%

C. Division Performance

Chocolate generated \$131,692.90 in sales and \$88,824.62 in gross profit, equivalent to 92.88% of company revenue and 95.06% of company gross profit. Its division gross margin is 67.45%. The fact that profit share exceeds revenue share indicates that Chocolate contributes disproportionately to profit and should be classified as the strongest financial division, even though an earlier phase report labeled it “Mixed / Benchmark

Performance.” This paper interprets the underlying metrics directly rather than repeating that label.

The Other division contributes 6.82% of revenue but only 4.64% of gross profit, with a margin of 44.84%. This negative revenue-to-profit conversion gap signals a structural margin problem. Sugar has a healthy aggregate margin of 66.61% but contributes only 0.30% of revenue and 0.30% of gross profit; therefore, it has little effect on total company economics despite several high-margin products.

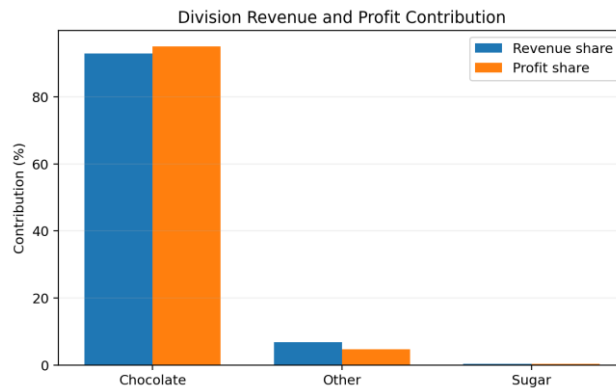


Fig. 4. Division revenue contribution compared with profit contribution.

TABLE V. DIVISION PERFORMANCE

Division	Sales (\$)	GP (\$)	Margin
Chocolate	131,692.90	88,824.62	67.45%
Other	9,663.25	4,333.45	44.84%
Sugar	427.48	284.73	66.61%

D. Product Concentration and Pareto Results

The revenue Pareto analysis shows that the top four products generate 76.26% of revenue, so a fifth product is required to cross the 80% threshold. Once the fifth product is included, the five core Chocolate products collectively account for 92.88% of revenue. The same structure appears in gross profit: four products account for 77.30% of profit and the fifth raises cumulative profit contribution to 95.06%. Thus, 5 of 15 products—33.33% of the portfolio—generate more than 90% of both revenue and profit.

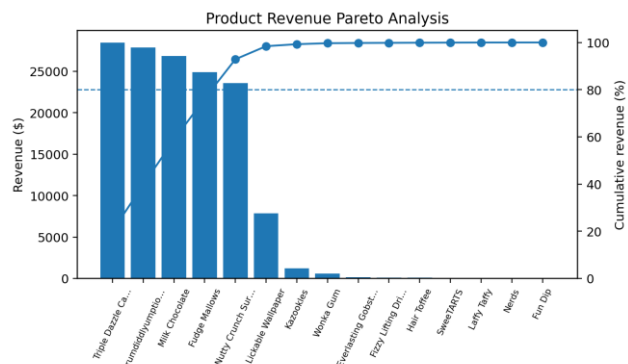


Fig. 5. Product revenue Pareto analysis; the dashed line marks the 80% cumulative threshold.

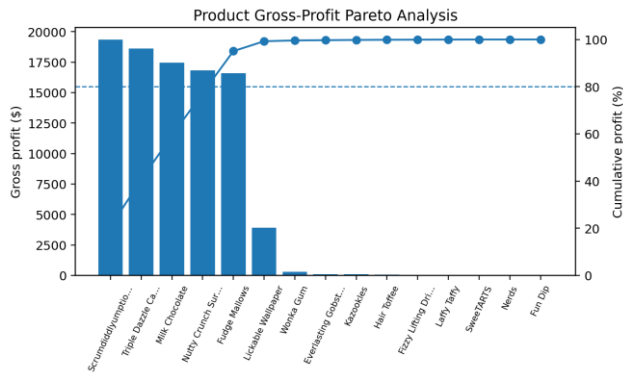


Fig. 6. Product gross-profit Pareto analysis; the same five Chocolate products dominate profit.

The concentration result is financially attractive in the sense that the dominant products have strong margins, but it introduces resilience risk. Any disruption affecting one or more of these Chocolate products—supplier cost increases, availability problems, price pressure, or demand decline—would have an outsized effect on the entire portfolio. Diversification should therefore be evaluated as a resilience objective rather than as an argument to reduce investment in the best-performing products.

E. Geographic Concentration

Geographic concentration is also material. Sixteen of 59 states/provinces, or 27.12%, are required to reach 80.03% of revenue, and the same number account for 80.14% of gross profit. California is the largest single market with 19.69% of revenue, 19.78% of profit, and 19.72% of orders. New York and Texas are the next major markets; together, California, New York, and Texas contribute approximately 40.11% of revenue. This concentration increases the business importance of demand, service quality, and commercial continuity in a relatively small set of markets.

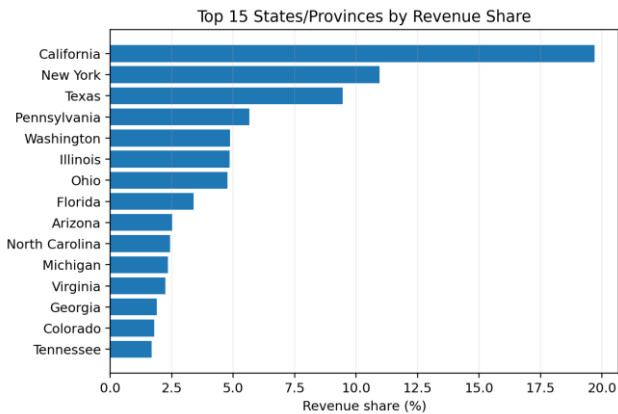


Fig. 7. Top 15 states/provinces by revenue share.

At the regional level, Pacific contributes 32.66% of revenue and 32.63% of profit, while Atlantic contributes 29.06% and 28.87%, respectively. Pacific and Atlantic together therefore account for 61.71% of revenue. Three of four regions are required to cross 80% cumulative revenue, reaching 84.31%. The project labels Pacific and Atlantic as high-load regions based on order and unit concentration. Because the dataset does not contain traffic, route capacity, warehouse throughput, or service-level measurements, these findings should be interpreted as demand/load concentration rather than proof of physical congestion.

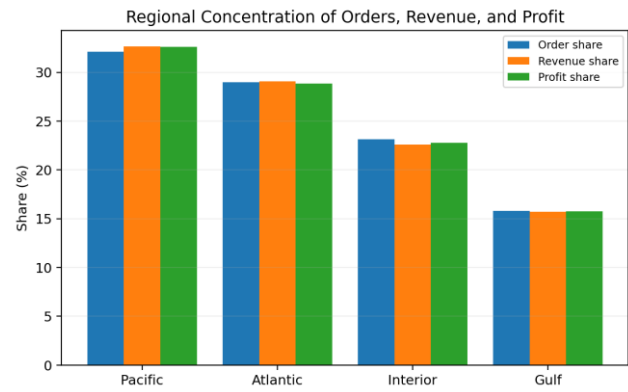


Fig. 8. Regional concentration of orders, revenue, and gross profit.

TABLE VI. GEOGRAPHIC DEPENDENCY SUMMARY

Indicator	Result
States/provinces required for 80% revenue	16 of 59 (27.12%)
Actual revenue share of those states	80.03%
States required for 80% profit	16
Actual profit share	80.14%
Top revenue state	California (19.69%)
Top region	Pacific (32.66% revenue)
Regions required for 80% revenue	3 of 4 (84.31%)

F. Cost Structure Diagnostics

Cost structure analysis separates high absolute cost from inefficient cost intensity. The five Chocolate products have the largest total costs because they dominate sales volume, yet they maintain gross margins from approximately 64.92% to 71.35%. Their high absolute cost is therefore scale-related rather than evidence of weak economics. Cost ratio, margin, sales scale, and profit must be considered together.

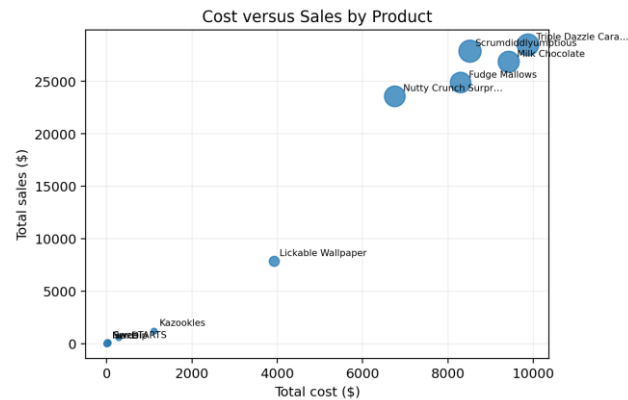


Fig. 9. Cost versus sales by product. Large Chocolate products have high absolute cost because of scale, not weak margin.

The quantile risk map identifies four products that are simultaneously cost-heavy and margin-poor: Kazookles, Fun Dip, SweetARTS, and Nerds. Kazookles is the most severe case, with \$1,113.00 of cost against \$1,205.75 of sales. Its cost ratio is 92.31%, gross margin 7.69%, and profit per unit only \$0.25. Because its sales exceed the median threshold of \$597.50, it is classified as a pricing inefficiency and a candidate for repricing as well as cost renegotiation.

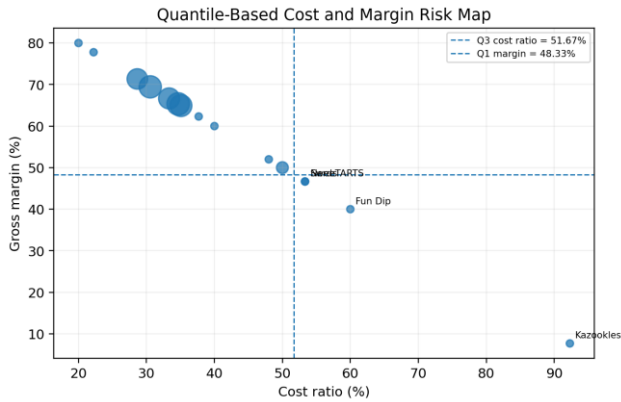


Fig. 10. Quantile-based cost and margin risk map. Dashed lines show the Q3 cost-ratio and Q1 margin thresholds.

Fun Dip, SweetTARTS, and Nerds meet a different pattern. Each is cost-heavy and margin-poor, and each also falls at or below the Q1 thresholds for sales and gross profit. Therefore, the model assigns them to discontinuation review. The terminology is deliberately “review” rather than “discontinue” because portfolio role, customer expectations, contractual constraints, and cross-selling effects are not represented in the dataset. Cost renegotiation is also appropriate for all four risk products because the common problem is a high share of revenue being consumed by cost.

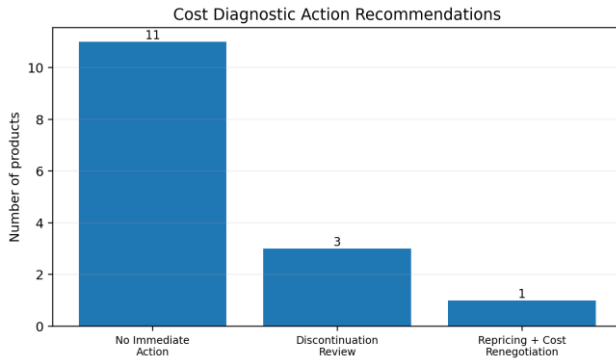


Fig. 11. Final action distribution from the quantile-based cost diagnostic rules.

TABLE VII. COST-DIAGNOSTIC PRIORITY PRODUCTS

Product	Cost ratio	Margin	Action
Kazookies	92.31%	7.69%	Reprice + renegotiate
Fun Dip	60.00%	40.00%	Discontinuation review
SweetTARTS	53.33%	46.67%	Discontinuation review
Nerds	53.33%	46.67%	Discontinuation review

G. Margin Volatility

The supplied margin-volatility summary reports an average company monthly margin of 65.91%, a standard deviation of 0.31 percentage points, a minimum of 65.39%, and a maximum of 66.35%. The corresponding range is 0.95 percentage points. This indicates a highly stable company-level margin in the supplied month-of-year aggregation. The result is consistent with the dataset structure, in which product-level price and cost relationships are largely fixed and overall variation is driven primarily by monthly changes in product mix.

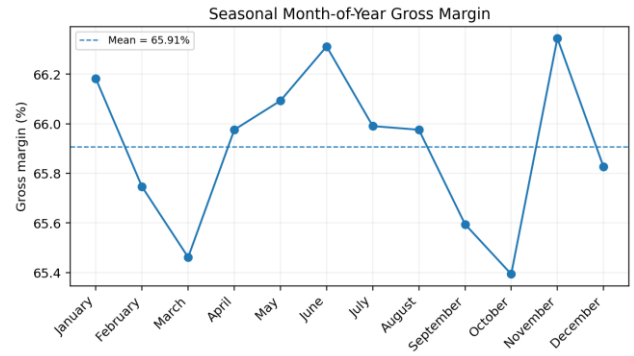


Fig. 12. Seasonal month-of-year company gross-margin pattern from the supplied volatility report.

Division-level volatility reveals a more uneven picture. Chocolate is highly stable, with a reported standard deviation of 0.16 percentage points. Other has a lower average monthly margin of 43.75% and a standard deviation of 2.53 percentage points, making it a structural concern because weak margin is combined with instability. Sugar has the highest volatility at 11.96 percentage points, with monthly margins ranging from 46.67% to 80.00%. However, Sugar is active in only 11 month-of-year buckets and represents only 0.30% of revenue, so its high volatility should be interpreted in the context of small scale.

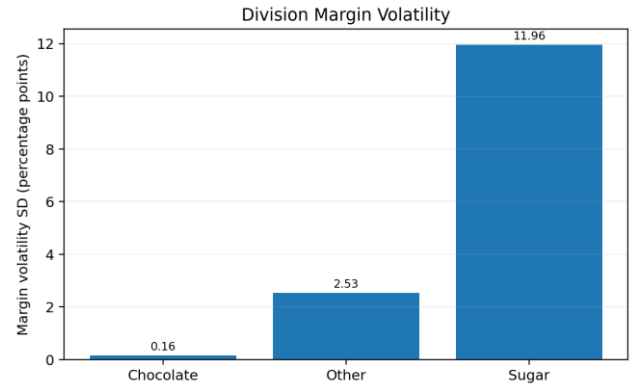


Fig. 13. Division margin volatility measured by standard deviation of the supplied monthly margin series.

TABLE VIII. DIVISION MARGIN VOLATILITY

Division	Avg. margin	SD	Range
Chocolate	67.43%	0.16 pp	0.58 pp
Other	43.75%	2.53 pp	7.35 pp
Sugar	62.59%	11.96 pp	33.33 pp

VI. STREAMLIT DECISION-SUPPORT DASHBOARD

A. Functional Architecture

The dashboard serves as the operational layer of the analysis. Its design follows the project technical document rather than introducing unrelated functionality. The Product Profitability module displays the margin leaderboard, profit contribution, and margin-risk view. Division Performance compares revenue and gross profit and presents margin distribution. Profit Concentration provides revenue and profit Pareto views plus state/region dependency indicators. Cost & Margin Diagnostics provides the cost-sales scatter, cost-ratio versus margin plot, quantile thresholds, and action flags. A separate Margin Volatility page adds company and division time-based views, while a Reports & Downloads page exposes the validated Colab outputs.

B. User Controls and Analytical Consistency

Four controls are central to the dashboard: a date-range selector, division filter, margin-threshold slider, and product search. Live profitability metrics are recalculated from the filtered row-level dataset. The margin-threshold slider is intentionally separated from the portfolio-relative cost action logic: it supports exploratory risk visualization but does not overwrite the approved Q1/Q3 action rules. This separation prevents an interactive display control from silently changing the classification methodology used in the research results.

C. Reproducibility and Deployment

The application is implemented in Python using pandas/NumPy for data transformation, Plotly for interactive charts, and Streamlit for the user interface [6], [8]. The deployment structure separates application code, requirements, and data files. Version control through GitHub allows the deployed application to update when validated code changes are pushed. In a production setting, confidential datasets should be stored in a private repository or external data service rather than exposed in a public project repository.

VII. DISCUSSION

A. Profit Quality versus Revenue Scale

The results confirm that revenue should not be interpreted as a standalone measure of product quality. Lickable Wallpaper, for example, generates \$7,860.00 in sales and \$3,930.00 in gross profit at a 50.00% margin; it is weaker than the Chocolate core but still creates meaningful profit. Kazookles generates far less profit from more than \$1,200 of sales because cost absorbs 92.31% of revenue. Conversely, Everlasting Gobstopper reports an 80% margin but only \$104 of gross profit. The decision framework therefore benefits from combining scale, margin, unit profit, and contribution.

B. Concentration as Both Strength and Risk

The same five Chocolate products dominate revenue and profit, and they do so with strong margins. This is a financially favorable form of concentration, because the business is not dependent on weak products. Nevertheless, the concentration creates strategic exposure. A portfolio in which one division contributes 95.06% of gross profit has limited financial insulation if the dominant product family experiences cost inflation or demand disruption. The appropriate response is not to reduce the successful core, but to protect its economics while developing additional profitable sources of contribution.

C. Relative Thresholds and Decision Transparency

Quantile-based thresholds improve internal consistency because the flags adapt to the observed portfolio distribution. They also make the logic auditable: stakeholders can see exactly why a product is considered high-cost or low-margin. However, relative thresholds can only compare products within the current portfolio. If every product were inefficient, quartiles would still label some products “better” than others. External benchmarks, target margins, supplier contract data, and competitor pricing would therefore strengthen future versions of the diagnostic model.

D. Stability and Product-Mix Effects

Company-level margin stability is largely a consequence of the dominance and stable economics of Chocolate. Product-level monthly margins are effectively constant, meaning that changes in company or division margins are driven primarily by mix. This is particularly evident in Sugar, where the small and irregular set of active products produces a wide aggregate margin range. The

finding is useful operationally because it suggests that margin volatility can be investigated through changes in product mix rather than only through price or cost changes.

VIII. MANAGERIAL IMPLICATIONS AND RECOMMENDATIONS

A. Protect the Chocolate Profit Engine

The five core Chocolate products should receive the highest priority for supply continuity, cost monitoring, and commercial availability because they jointly contribute 95.06% of gross profit. Management should monitor supplier changes and cost increases closely, avoid unnecessary discounting that would erode strong margins, and maintain service levels in the largest geographic markets. Scenario analysis should estimate the company-wide impact of a 1–5 percentage-point margin decline in these products because even a small deterioration would materially affect total profit.

B. Correct Kazookles Economics

Kazookles requires a focused commercial and sourcing review. The immediate objective is to increase the price-cost gap through repricing, cost renegotiation, or both. Because its sales exceed the median product sales threshold, discontinuation should not be the first action. A practical management test is to simulate target margins (for example, the portfolio Q1 margin threshold and the Other-division average) and calculate the required price increase or cost reduction. If neither is commercially feasible, a later rationalization decision may be justified.

C. Review Low-Scale, Low-Profit Sugar Products

Fun Dip, SweetARTS, and Nerds are candidates for discontinuation review because their weak margins are accompanied by very low sales and gross profit. Management should first check whether these products play a strategic role in assortment completeness, customer acquisition, or bundled orders. If no such role is established, removing or renegotiating these products can simplify the portfolio without sacrificing meaningful profit.

D. Reduce Geographic and Division Dependency

California and the Pacific region warrant special attention because they represent the largest geographic shares, but the objective should be resilience rather than volume reduction. Management should protect service quality in these markets while seeking profitable growth in less concentrated regions. At the division level, the business should explore ways to develop non-Chocolate products with sustainable margins so that overall profit becomes less dependent on a single product family.

E. Establish a Recurring Review Cycle

The dashboard should be used as a recurring management tool rather than as a one-time project output. A monthly or quarterly review can track shifts in margin, cost ratio, contribution, concentration, and action flags. Quantile thresholds should be recalculated only under a documented policy, because frequent recalculation can cause products to move between categories simply because the portfolio composition changes. Baseline thresholds and current-period thresholds should therefore be shown together when the dashboard is used for governance.

IX. LIMITATIONS AND THREATS TO VALIDITY

Several limitations constrain the interpretation of the findings. First, the dataset contains no external market prices, competitor benchmarks, supplier contract terms, promotional spend, customer acquisition costs, or overhead allocations. Gross profit is therefore a product-level contribution measure rather than a

complete measure of net profitability. Second, the cost-diagnostic thresholds are internal quartiles and should not be described as industry standards. Third, geographic “high load” is inferred from orders and units; actual congestion cannot be established without logistics capacity, route, traffic, and service-level data.

Fourth, the archived cleaned dataset contains implausible ship-date and shipping-duration fields even though quality flags mark them valid. These fields are excluded from the substantive findings. Fifth, the supplied margin-volatility report collapses the two-year period into 12 month-of-year buckets; it therefore measures seasonal month-of-year variation rather than 24-point chronological volatility. Sixth, product-level margins are nearly constant by construction, limiting the ability to study price and cost shocks. Finally, the dataset contains only 15 products, so quartile thresholds can be sensitive to individual product values.

These limitations do not invalidate the profitability and concentration calculations, which reconcile across phase reports, but they define the scope within which the business recommendations should be applied. Future work should incorporate supplier price histories, discount/promotion data, net operating costs, customer segments, inventory availability, and corrected fulfillment dates.

X. REPRODUCIBILITY AND IMPLEMENTATION NOTES

The project is designed to be reproducible from the cleaned transaction dataset. Product, division, Pareto, geographic, cost, and volatility outputs are stored as CSV reports. The research paper uses those outputs as the primary validated evidence and uses the cleaned row-level data for profile checks and dashboard filtering. The code base should retain a requirements file with pinned or documented versions of Streamlit, pandas, NumPy, and Plotly for deployment. A Git repository should exclude virtual environments, cache directories, and secret files through .gitignore.

For research reproducibility, the strongest practice is to preserve: (1) the original raw dataset; (2) the cleaned dataset; (3) data-quality reports; (4) scripts/notebooks used for each phase; (5) exported phase outputs; (6) the dashboard application; and (7) a README documenting the run order and known data limitations. This separation allows future reviewers to distinguish raw evidence from transformed analytical outputs and presentation artifacts.

XI. CONCLUSION

This study demonstrates a practical, transparent framework for product-line profitability and margin-risk analysis using the Nassau Candy Distributor project. The company is strongly profitable in aggregate, generating \$93,442.80 of gross profit from \$141,783.63 of sales at a 65.91% gross margin. However, profitability is structurally concentrated: five Chocolate products produce 92.88% of revenue and 95.06% of gross profit. Chocolate is therefore both the primary economic engine and the primary source of portfolio dependency.

The cost diagnostic adds decision specificity to the EDA. Quantile-based thresholds identify Kazookles as the clearest repricing and cost-renegotiation priority, while Fun Dip, SweetARTS, and Nerds merit discontinuation review because low scale and low profit coincide with cost and margin risk. Geographic analysis shows a second layer of concentration, with 16 of 59 states/provinces generating approximately 80% of both revenue and profit. Margin analysis indicates strong company-

level stability in the supplied seasonal monthly aggregation, but Sugar and Other are less stable than Chocolate.

The broader contribution is methodological: portfolio decisions become more defensible when sales, margin, contribution, cost intensity, concentration, and volatility are considered together and exposed through an interactive dashboard. Future work should incorporate corrected shipping data, supplier/market benchmarks, promotions, net profitability, and richer time-varying price-cost data. With those extensions, the framework could evolve from descriptive and diagnostic analytics toward predictive margin-risk monitoring and prescriptive product-portfolio optimization.

XII. RESEARCH QUESTION SYNTHESIS

TABLE IX. RESEARCH QUESTIONS AND FINDINGS

RQ	Evidence-based answer
RQ1	Core Chocolate products dominate gross profit; high sales do not always imply strong margin, as shown by Kazookles.
RQ2	Five products generate 92.88% of revenue/profit; 16 of 59 states generate ~80%; Pacific and Atlantic dominate regional load.
RQ3	Quantile rules identify four simultaneous cost-heavy/margin-poor products.
RQ4	Kazookles: repricing + cost renegotiation; Fun Dip, SweetARTS, Nerds: discontinuation review.
RQ5	Company seasonal month-of-year margin SD = 0.31 pp; Chocolate 0.16 pp; Other 2.53 pp; Sugar 11.96 pp.
RQ6	Streamlit operationalizes filters, product/division views, Pareto, cost diagnostics, volatility, and report downloads.

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APPENDIX A. COMPLETE PRODUCT PROFITABILITY AND ACTION MAP

Table X consolidates the final portfolio metrics and recommended action flags used in the paper. Values are taken from the validated Phase 3.2 and Cost Structure Diagnostics outputs. “No immediate action” does not imply that a product requires no management attention; it means that the product does not meet the specific cost-diagnostic intervention rules used in this study.

TABLE X. FULL PRODUCT PROFITABILITY AND COST-ACTION MAP

Product	Div.	Sales \$	GP \$	Margin	Cost ratio	Final action
Wonka Bar - Scrumdiddlyumptious	Chocolate	27,874.80	19,357.50	69.44%	30.56%	No Immediate Action
Wonka Bar - Triple Dazzle Caramel	Chocolate	28,485.00	18,610.20	65.33%	34.67%	No Immediate Action
Wonka Bar - Milk Chocolate	Chocolate	26,867.75	17,443.37	64.92%	35.08%	No Immediate Action
Wonka Bar - Nutty Crunch Surprise	Chocolate	23,574.95	16,819.95	71.35%	28.65%	No Immediate Action
Wonka Bar - Fudge Mallows	Chocolate	24,890.40	16,593.60	66.67%	33.33%	No Immediate Action
Lickable Wallpaper	Other	7,860.00	3,930.00	50.00%	50.00%	No Immediate Action
Wonka Gum	Other	597.50	310.70	52.00%	48.00%	No Immediate Action
Everlasting Gobstopper	Sugar	130.00	104.00	80.00%	20.00%	No Immediate Action
Kazookles	Other	1,205.75	92.75	7.69%	92.31%	Repricing + Cost Renegotiation
Hair Toffee	Sugar	76.50	59.50	77.78%	22.22%	No Immediate Action
Fizzy Lifting Drinks	Sugar	78.75	47.25	60.00%	40.00%	No Immediate Action
Laffy Taffy	Sugar	53.73	33.48	62.31%	37.69%	No Immediate Action
SweeTARTS	Sugar	61.50	28.70	46.67%	53.33%	Discontinuation Review
Nerds	Sugar	15.00	7.00	46.67%	53.33%	Discontinuation Review
Fun Dip	Sugar	12.00	4.80	40.00%	60.00%	Discontinuation Review

APPENDIX B. KEY FORMULAS AND DECISION RULES

TABLE XI. FORMULAS AND FINAL COST-DIAGNOSTIC RULES

Metric / rule	Definition
Gross margin (%)	$\text{Gross Profit} / \text{Sales} \times 100$
Profit per unit	$\text{Gross Profit} / \text{Units}$
Revenue contribution (%)	$\text{Product Sales} / \text{Total Sales} \times 100$
Profit contribution (%)	$\text{Product Gross Profit} / \text{Total Gross Profit} \times 100$
Cost ratio (%)	$\text{Cost} / \text{Sales} \times 100$
Low margin	$\text{Gross Margin} \leq \text{portfolio Q1 (48.33\%)}$
High cost	$\text{Cost Ratio} \geq \text{portfolio Q3 (51.67\%)}$
Pricing inefficiency	High cost AND low margin AND Sales $\geq$ median (\$597.50)
Cost renegotiation	High cost AND low margin
Discontinuation review	High cost AND low margin AND Sales $\leq$ Q1 (\$69.00) AND Gross Profit $\leq$ Q1 (\$40.36)

APPENDIX C. DATA INTEGRITY NOTES FOR SUBMISSION

Before formal conference or journal submission, the author should replace the affiliation and email placeholders in the author block, check the target venue page limit, and apply the venue-specific IEEE template if required. The shipping-date anomaly should remain disclosed unless the underlying dataset is corrected and the affected analyses are rerun. The supplied margin-volatility report should also be described explicitly as a 12-bucket month-of-year aggregation across the two-year order period. These disclosures preserve the distinction between validated project findings and fields that are not reliable enough for inference.

APPENDIX D. STATE/PROVINCE CONCENTRATION DETAIL

The following table preserves the state/province-level concentration output used to derive the geographic dependency findings. It is included for auditability and reproducibility; the main paper reports only the most decision-relevant concentration statistics.

TABLE XII. COMPLETE STATE/PROVINCE CONCENTRATION OUTPUT

State / Province	Orders	Sales \$	Rev. share	Profit share	Load flag
California	1686	27,917.40	19.69%	19.78%	High Load
New York	942	15,541.03	10.96%	10.94%	High Load
Texas	833	13,416.09	9.46%	9.53%	High Load
Pennsylvania	487	8,027.03	5.66%	5.59%	High Load
Washington	432	6,921.15	4.88%	4.89%	High Load
Illinois	433	6,898.96	4.87%	4.88%	High Load
Ohio	392	6,768.95	4.77%	4.72%	Normal Load
Florida	319	4,804.02	3.39%	3.43%	Normal Load
Arizona	190	3,587.55	2.53%	2.45%	Normal Load
North Carolina	216	3,450.86	2.43%	2.49%	Normal Load
Michigan	209	3,331.00	2.35%	2.39%	Normal Load
Virginia	185	3,177.84	2.24%	2.23%	Normal Load
Georgia	151	2,692.84	1.90%	1.87%	Normal Load
Colorado	140	2,544.91	1.79%	1.81%	Normal Load
Tennessee	151	2,383.56	1.68%	1.68%	Normal Load
Indiana	123	2,002.78	1.41%	1.44%	Normal Load
Wisconsin	91	1,920.50	1.35%	1.31%	Normal Load
Oregon	105	1,920.15	1.35%	1.34%	Normal Load
Kentucky	112	1,880.63	1.33%	1.33%	Normal Load
Massachusetts	106	1,806.99	1.27%	1.26%	Normal Load
New Jersey	108	1,578.35	1.11%	1.09%	Normal Load
Maryland	80	1,476.99	1.04%	1.07%	Normal Load
Delaware	80	1,323.59	0.93%	0.92%	Normal Load
Minnesota	75	1,161.46	0.82%	0.83%	Normal Load
Alabama	53	995.11	0.70%	0.67%	Normal Load
Connecticut	74	980.57	0.69%	0.70%	Normal Load
Arkansas	48	891.94	0.63%	0.63%	Normal Load
Missouri	56	882.83	0.62%	0.64%	Normal Load
Oklahoma	56	877.65	0.62%	0.64%	Normal Load
Utah	44	856.41	0.60%	0.60%	Normal Load
Mississippi	43	844.18	0.60%	0.60%	Normal Load
Ontario	41	814.57	0.57%	0.58%	Normal Load
Rhode Island	46	805.76	0.57%	0.54%	Normal Load
Nevada	36	759.46	0.54%	0.50%	Normal Load
South Carolina	34	601.07	0.42%	0.43%	Normal Load
Quebec	38	597.68	0.42%	0.41%	Normal Load
Alberta	22	530.32	0.37%	0.35%	Normal Load
Louisiana	38	525.21	0.37%	0.37%	Normal Load
New Mexico	34	502.71	0.35%	0.36%	Normal Load
Nebraska	33	483.53	0.34%	0.35%	Normal Load
New Hampshire	26	428.27	0.30%	0.28%	Normal Load
Iowa	26	400.37	0.28%	0.29%	Normal Load
British Columbia	22	290.90	0.21%	0.21%	Normal Load
Kansas	23	263.18	0.19%	0.19%	Normal Load

State / Province	Orders	Sales \$	Rev. share	Profit share	Load flag
Idaho	18	225.25	0.16%	0.16%	Normal Load
Montana	12	202.11	0.14%	0.14%	Normal Load
New Brunswick	8	172.72	0.12%	0.13%	Normal Load
Vermont	10	162.44	0.11%	0.12%	Normal Load
Newfoundland and Labrador	5	162.00	0.11%	0.12%	Normal Load
South Dakota	8	151.34	0.11%	0.11%	Normal Load
District of Columbia	8	142.25	0.10%	0.10%	Normal Load
Manitoba	8	138.25	0.10%	0.10%	Normal Load
Prince Edward Island	8	130.10	0.09%	0.09%	Normal Load
Maine	7	126.48	0.09%	0.09%	Normal Load
North Dakota	5	109.66	0.08%	0.08%	Normal Load
Nova Scotia	6	87.50	0.06%	0.06%	Normal Load
West Virginia	4	63.97	0.05%	0.05%	Normal Load
Saskatchewan	2	29.25	0.02%	0.02%	Normal Load
Wyoming	1	13.96	0.01%	0.01%	Normal Load

APPENDIX E. MARGIN VOLATILITY DETAIL

Product-level volatility is effectively zero in the supplied report because each product retains a fixed sales-cost relationship. The table below documents the product-level values and active month-of-year buckets.

TABLE XIII. PRODUCT MARGIN VOLATILITY OUTPUT

Product	Division	Active	Avg margin	SD	Category
Wonka Bar - Fudge Mallows	Chocolate	12	66.67%	0.00	Low Volatility
Wonka Bar - Milk Chocolate	Chocolate	12	64.92%	0.00	Low Volatility
Wonka Bar - Nutty Crunch Surprise	Chocolate	12	71.35%	0.00	Low Volatility
Wonka Bar - Triple Dazzle Caramel	Chocolate	12	65.33%	0.00	Low Volatility
Wonka Bar - Scrumdiddlyumptious	Chocolate	12	69.44%	0.00	Low Volatility
Kazookles	Other	12	7.69%	0.00	Low Volatility
Lickable Wallpaper	Other	12	50.00%	0.00	Low Volatility
Wonka Gum	Other	12	52.00%	0.00	Low Volatility
Everlasting Gobstopper	Sugar	3	80.00%	0.00	Low Volatility
Fizzy Lifting Drinks	Sugar	5	60.00%	0.00	Low Volatility
Fun Dip	Sugar	2	40.00%	0.00	Low Volatility
Hair Toffee	Sugar	3	77.78%	0.00	Low Volatility
Laffy Taffy	Sugar	6	62.31%	0.00	Low Volatility
Nerds	Sugar	3	46.67%	0.00	Low Volatility
SweeTARTS	Sugar	7	46.67%	0.00	Low Volatility

TABLE XIV. SUPPLIED COMPANY MONTH-OF-YEAR MARGIN REPORT

Month bucket	Sales \$	Gross profit \$	Margin
April	8,910.66	5,878.91	65.98%
August	10,217.75	6,741.27	65.98%
December	21,234.48	13,977.92	65.83%
February	3,933.52	2,586.14	65.75%
January	5,583.00	3,694.95	66.18%
July	9,923.05	6,548.31	65.99%
June	9,733.69	6,454.61	66.31%
March	9,562.14	6,259.54	65.46%
May	10,507.20	6,944.54	66.09%
November	20,646.57	13,697.97	66.35%
October	12,161.87	7,953.11	65.39%
September	19,369.70	12,705.53	65.59%

APPENDIX F. ANALYTICAL OUTPUT LINEAGE AND DASHBOARD MAPPING

TABLE XV. PROJECT ANALYTICAL LINEAGE

Stage	Analytical purpose	Primary output
Phase 1	Data understanding	Rows, columns, entities, total sales/units/profit/cost
Phase 2	Cleaning & validation	Field types, missingness, quality flags, known shipping anomaly
Phase 3.1	Profitability metrics	Margin, profit/unit, profit contribution
Phase 3.2	Product analysis	Ranks and profitability categories
Phase 3.3	Division analysis	Revenue/profit shares and division margins
Phase 3.4	Concentration	Product Pareto, state/region dependency
Cost diagnostics	Portfolio-relative risk	Q1/Q3 cost/margin flags and recommended actions
Volatility	Margin stability	Company, division, and product variability
Streamlit	Decision support	Filters, live charts, risk flags, report downloads

Dashboard mapping. Product Profitability uses the cleaned transaction file for live filtering and Phase 3.2 as a validated baseline. Division Performance uses row-level aggregation and the Phase 3.3 report. Profit Concentration uses live product aggregation plus saved Pareto and geographic reports. Cost & Margin Diagnostics preserves the final quantile-based action rules from Cost_Structure_Diagnostics.csv. Margin Volatility uses Order Date and the supplied volatility reports. Reports & Downloads exposes the archived Colab CSV outputs so that dashboard views can be compared with the validated analysis.